

ELEN4730

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1 Lecture 1/19/24

1.1 Quantum State Qubit

There are 3 steps of a quantum computer:

1. state preparation
2. quantum operator
3. measurement

Postulate 1 For an isolated system at any time, it is completely specified by a state $|\psi\rangle \in \mathcal{H}$ where the absolute value is equal to 1. For a qubit, it exists such that $\mathcal{H} = C^2$ and is represented in the computation basis as $|0\rangle$ and $|1\rangle$. Thus, qubits will be in the states $\alpha|0\rangle + \beta|1\rangle$. Now if two qubits differ by a local phase factor, we understand that they are both solutions of the same wave equation and thus are going to be equivalent eigenvalues.

1.2 Bloch Sphere Representation

Consider $\alpha = r_0 e^{j\theta_0}, \beta = r_1 e^{j\theta_1} \implies r_0^2 + r_1^2 = 1$. This implies that we can represent $|\psi\rangle = r_0 e^{j\theta_0} |0\rangle + r_1 e^{j\theta_1} |1\rangle = \cos(\theta/2) |0\rangle + \sin(\theta/2) e^{j\phi} |1\rangle, \theta \geq \phi \geq 2\pi$ where on the bloch sphere, θ represents the azimuthal angle, and ϕ represents the rotation angle. We can then denote the x, z, and y axis for different computation basis.

1.3 Quantum Operators

Postulate Closed systems evolve according to a unitary operator. Now, if we consider an open system, the system doesn't evolve according to a unitary operator. Rather, it evolves with respect to the the Linbald Master Equation. Unitary operators obey the following properties:

$$UU^\dagger = I \implies U^\dagger = U^{-1}$$

Now recall that the eigenvalue of a unitary operator is going to be of the form $\lambda = e^{i\theta}$. Now for qubits and qubit gates, they are represented as unitary operators given in the computational basis usually. Thus, we can consider the following unitary operators:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} Z = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} H = 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

From here, we can define the following pauli operators which are the set $[I, X, Y, Z]$ where the Y gate is a rotation in the imaginary direction with

$$Y = \begin{bmatrix} 0 & -j \\ j & 0 \end{bmatrix}$$

If we consider an eigendecomposition of the following pauli operators, we will get that I, X, Z, and Y will be:

$$I = |v_0\rangle\langle v_0| + |v_1\rangle\langle v_1| X = |0\rangle\langle +| - |-\rangle\langle -| Z = |0\rangle\langle 0| - |1\rangle\langle 1| Y = |j\rangle\langle j| - |-j\rangle\langle -j|$$

Through this representation and the bloch sphere, we know that there are cyclic properties in which multiply by certain operators will result in different gates. Additionally, all the Pauli gates exhibit anti-commutative properties.

1.3.1 Pauli Rotation Operators

Recall that for any unitary operator, we can represent it in the form of $U = e^{-jG}$ where G is defined to be a Hermitian operator. That is,

$$G = \sum_{i=1} \lambda_i |\lambda_i\rangle\langle \lambda_i|$$

Now let us reconsider the Pauli Operators as Hermitian operators, that is, they are self adjoint. Therefore we will have that we can represent the Paul operators as

$$R(\theta) = \cos(\theta/2) - j\sin(\theta/2)Q, Q = |v_0\rangle\langle v_0| - |v_1\rangle\langle v_0|$$

. Since we have defined this rotation on the bloch sphere which representations possible gates. We can now consider the special cases for the rotation operator:

$$R_Q(\pi) = Q, Q \in [X, Y, Z], R_Z(\pi/2) = S, R_z(\pi/4) = T$$

1.4 Quantum Measurement

Postulate 3 Von Neumann Measurement: Given an orthonormal basis of vectors, the system state ψ , the state before measurement, will have a measurement outcome for a state $m = i$ with $P = ||b_i\rangle\langle\psi||^2$. If the operator is $m = i$, then the post state will be represented s.t. $|\phi_{post}^{(i)}\rangle = |b_i\rangle$. One example is the computation basis which can be represented as $|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$. Taking the measurement we can represent it as $P_0 = ||0\rangle\langle\phi||^2$

Now suppose we took a measurement in the computational basis, the measurement output would be 0 or 1. Now for $\forall U = [|U_0\rangle, |U_1\rangle]I$, we will have that $U = |U_0\rangle\langle 0| + |U_1\rangle\langle 1|$ which is a change of basis to the U-basis. Thus, we can perform measurements in the U -basis. Now if we want to change back the the computational basis, we simply apply the conjugate transpose of the U operator since all operators are Hermitian.

1.4.1 Data Encoding

Consider the state $|\phi_x\rangle$ with n qubits represented by a $n \times 1$ matrix. Now suppose we had a 2×1 matrix that we denote as $\mathbf{x}$. We can represent the state $\mathbf{x}$ as a four dimensional vector which we can denote as a feature map, i.e. a mapping to a higher dimensional state space. Why would we perform basis encoding exactly? IT does well in the basis states, it allows us to change values more easily, and we can perform angle encoding much more easily

2 Lecture 1/26/24

Let us again consider a qubit in hilbert space $\mathcal{H}$. The state is represented in an orthonormal basis in which the coefficients of that basis are complex numbers. Again, let us consider the unitary operator U and perform U the computational basis. Now suppose that U is the Hadamard gate, applied to our computational basis we will get the Hadamard basis. Again, for different values of $U = [|u_0\rangle|u_1\rangle]$ we can create certain gates where $U \in \mathbb{C}^{2 \times 2}$. Thus we can represent all Pauli Operators, Pauli State Gates, and Rotation Gates in the form of Unitary Operators.

Now let us consider the Von Neumann Measurement using the unitary operator U . We can measure with respect to our U basis which will give us our probabilities of measurement in the u basis.

2.1 Multiple Qubits

Let us consider two qubits with states $|x\rangle$ and $|y\rangle$. The state of these qubits together will be represented as a tensor product s.t.

$$|x\rangle \otimes |y\rangle$$

which is a $mn \times 1$ matrix. Now let us consider to matrices s.t. $A \in \mathbb{C}^{m \times n}, B \in \mathbb{C}^{p \times q}$. Thus $A \otimes B$ is a $mp \times nq$ matrix. Now from thus, we will define the following tensor properties:

1. $(A \otimes B)^T = A^T \otimes B^T$
2. $(A \otimes B)^* = A^* \otimes B^*$
3. $(A \otimes B)(C \otimes D) = AC \otimes BD$

For Hermitian A and B , we will have that $A^* = A, B^* = B$

Tensor Product Space Given a two qubit space we can define the tensor product space $\mathcal{H} = \mathcal{H}_\infty \otimes \mathcal{H}_\epsilon$ which can be represented as

$$\sum_i \alpha_i |x_i\rangle \otimes \langle y_i|$$

. This tensor space will be defined by the basis vectors of each individual space tensored together.

2.2 Multi Qubit Systems

Posulate 4: A system composed of two subsystems defined by two hilbert spaces must be treated as a tensored system of the two hilbert spaces. We can write a state $|\varphi\rangle = |\varphi_1\rangle \otimes |\varphi_2\rangle$.

Let us now consider multi qubit gates. Firstly, consider the Hadamard gate which is represented as

$$H^{\otimes n \times n}, |H\rangle^n |0\rangle_n = \frac{1}{\sqrt{2}} \sum_{i=0}^{2^n-1} |i\rangle_n$$

2.3 Control U-Gate

Consider a control gate in which there exists a target and control qubit which we will write as $|q_1 q_0\rangle$. The gate will perform such that

1. $CU|00\rangle = |00\rangle$
2. $CU|01\rangle = U|0\rangle \otimes |1\rangle = (0, u_{00}, 0, u_{10})$
3. $CU|10\rangle = |10\rangle$
4. $CU|11\rangle = U|1\rangle \otimes |0\rangle = (0, u_{01}, 0, u_{11})$

Therefore we can define the CU gate as

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & u_{00} & 0 & u_{01} \\ 0 & 0 & 1 & 0 \\ 0 & u_{10} & 0 & u_{11} \end{pmatrix}$$

With this representation, we can consider multiple different gates which can be referred to as CX or the $CNOT$ gate, where $U = X$. There are other gates that we can consider such as the SWAP gate which all contribute to the different things needed to do.

2.4 Multi Qubit Measurement

Recall once again Von Neumann Measurement which has an orthonormal basis with N basis vectors with N different outputs. For $P(m = x) = |\langle u_x | \varphi \rangle|^2$ which is our post measurement state.

Partial Measurement Let us consider a separable system where we can represent $\varphi = |q_1\rangle \otimes |q_0\rangle$ where our goal is to measure q_0 only.

Now we know that

$$\begin{aligned} P(m=0) &= |\langle \varphi | I \otimes |0\rangle \langle 0| | \varphi \rangle|^2 = |\langle q_1 | \otimes \langle q_0 | I \otimes |0\rangle \langle 0| | q_1 \rangle \otimes |q_0\rangle|^2 \\ &= |\langle q_1 | I | q_1 \rangle|^2 |\langle q_0 | 0 \rangle \langle 0 | q_0 \rangle|^2 \end{aligned}$$

Therefore we will have that $P(m=0) = |\langle \varphi | I \otimes |0\rangle \langle 0| | \varphi \rangle|^2$ (I am not sure if the math here is right). But we will result in

$$|\varphi_{\text{post}}\rangle = \frac{I \otimes |0\rangle \langle 0| \varphi}{\sqrt{P(m=0)}}$$

3 Lecture 2/2/24

Consider a single qubit in the computational basis, we understand that $\mathcal{H} = \mathbb{C}^2$. For a two qubit system, we understand that we will get four different states that can be represented as

$$|00\rangle, |10\rangle, |11\rangle, |01\rangle$$

From here we can thus represent the state $\psi = \sum_{i=0}^3 a_i |i\rangle$.

Note: We can have entangled and separable states in which entangled states can not be written as the tensor product between two states.

Now suppose we had a separable operator where $U = U_1 \otimes U_2$. We will get that

$$U|\psi\rangle = \sum a_K U|k\rangle$$

which simply is a change of basis based on the operator which is unitary. Continuing, we will now consider a system with n different qubits where which is represented by the computational basis $|0\rangle, |1\rangle, \dots, |N-1\rangle$. And for these, we can consider unitary operators which act on all these gates in the same way. For example, we can take the Hadamard Gate and apply it to all the computational basis.

Universal Gate and No Cloning Theorem The set of gates CNOT, H, T combined can create any other unitary operations not found. Additionally, we have the no-cloning theorem which simply states that a universal copier of quantum states doesn't exist, that is

$$U|\psi\rangle \otimes |\phi\rangle = |\psi\rangle \otimes |\phi\rangle, \forall \phi, \psi \in \mathcal{H}$$

3.1 Projectors and Projective measurements

We will note the following the following definitions:

1. idempotent: $\Pi^2 = \Pi$
2. Hermitian $\Pi = \Pi^T$ and (1)

Projective measurements are thus going to be $\sum_{i=1}^m \Pi_i = I$. Thus we understand that the sum of the projective measurements of each state i will result in the value 1.

Now consider a system $X, Pr(k) = ||\Pi_k^2|\psi\rangle||^2$. $X = \frac{\Pi_k|\psi\rangle}{||\Pi_k|\psi\rangle|}$. Now let us consider a system of (X, Y) which becomes $Pr(a) = ||\Pi_a \oplus I|\psi\rangle||^2$ Thus we will get that

$$\psi = \frac{(\Pi_a \oplus I)|\psi\rangle}{||\Pi_a \oplus I|\psi\rangle||}$$

A projective measurement on ψ consists of projectors $P_0, P_1, \dots, P_{M-1}$ s.t. $\sum_{k=0}^{M-1} P_k = I_N$ with outcome $m = k, pr(k) = \langle\psi|P_k|\psi\rangle$. Our post-measurement state will be represented as

$$|\psi_{post}^{(k)}\rangle = \frac{P_k|\psi\rangle}{\sqrt{P_k}}$$

Example: For a 2-qubit system where $N = 5, M = 4$ and $P_k = |k\rangle\langle k|, k = 0, 1, 2, 3$ we know that $pr(k) = \langle\psi||k\rangle\langle k||\psi\rangle = |a_k|^2, |\psi\rangle = \sum a_k|k\rangle$. Thus we will have that

$$|\psi_{post}^{(k)}\rangle = e^{i\theta}|k\rangle = |k\rangle$$

For partial measurements we will have the form

$$|\psi_{post}^{(0)}\rangle = \frac{a_0|00\rangle + a_{10}|10\rangle}{\sqrt{a_{00}^2 + a_{10}^2}}$$

3.2 Parity Measurement

Consider a system with $M = 2$ we will have that

$$\bar{P}_0 = |00\rangle\langle 00| + |11\rangle\langle 11|, \bar{P}_1 = |01\rangle\langle 01| + |10\rangle\langle 10|$$

To implement this parity measurement we will have that

3.3 Quantum Teleportation

We will consider the protocol in how this happens. Let us have Alice and Bob share the aforementioned state ϕ^+ . They will be able to share a c-bit and transfer information.

4 Lecture 2/23/24

Consider the following

$$|\psi\rangle_d = H^{\otimes n}|x_o\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in [0,1]} (-1)^{x \cdot y} |y\rangle$$

From here, we get a final measurement of $P(y) = \frac{1}{2^n}$. If $r = 0$, the final measurement is uniform in $\{0, 1\}^n$. I

5 Lecture 3/22/24

We consider Binary Quadratic Optimization in the form of the following:

$$\min g(x) = \sum_{i,j} a_{ij}x_i x_j + \sum_i b_i x_i$$

There exists two approaches to the problem:

5.0.1 Approach 1: QUBO

QUBO problems have $x \in \{0, 1\}$ where we recursively search for x s.t. $g(x) < g(x_0)$ for fixed x_0 which is called Grover's Adaptive Search (GAS)

5.0.2 Approach 2: Ising Model $x \in \{-1, +1\}$

We use the Ising Model in this approach and analyze the Hamiltonian of a problem with the form:

$$H = \sum a_{ij} z_{ij} + b_i z_i$$

where the Pauli String: $Z_{ij} = I \otimes \dots \otimes I \otimes Z_i \otimes \dots \otimes Z_j \otimes \dots \otimes I$. In this case we would like to minimize the problem s.t.

$$\min g(x) \equiv \min \langle x | H | x \rangle = \min \langle \psi | H | \psi \rangle$$

where ψ is in any state in $\mathbb{C}^{2^n}$

6 Adiabatic QC

Quantum state evolves continuously, according to the time-dependent Schrodinger equation:

$$H(t)|\psi\rangle = i\hbar \partial_t |\psi\rangle$$

Where $H(t)$ is a time dependent Hamiltonian. We wish to minimize the energy to the ground state s.t. $|\psi^*\rangle = \arg \min \langle \psi | H | \psi \rangle$

6.1 Basic Idea

Start with a simple Hamiltonian for which the ground state can be easily prepared. Slowly change $H(t)$ s.t. the state $|\psi\rangle$ remains at the ground state at all time. The wave function changes s.t. it remains at the ground state. The end Hamiltonian is H_1 corresponding to our problem the final solution.

Note: The exact methodology to build the Hamiltonian for continuous ground state is unknown. The use of the D-wave machine and quantum annealing allows us to build such a machine.

Total Time: T , initial: H_0 end: H_1

$$\implies H(t) = A(t)H_0 + B(t)H_1$$

$$A(0) = 1, A(T) = 0, B(0) = 0, B(T) = 1 \text{ e.g. } A(t) = 1 - t/T, B(t) = t/T$$

6.2 Quantum Annealing

Initial Hamiltonian:

$$H_0 = - \sum_{i=0}^{n-1} X_i, \text{ where } X_i = I \otimes I \dots \otimes X_i \otimes \dots \otimes I$$

Then $|\psi_0\rangle = |+\rangle^{\otimes n}$ is a ground state of H_0

6.2.1 Proof

$$X = |+\rangle\langle+| - |-\rangle\langle-|$$

$$I = |+\rangle\langle+| + |-\rangle\langle-|$$

$$|U_x\rangle = \{|+\rangle, |-\rangle\}^{\otimes n} \text{ forms a basis of n-qubits}$$

$$|\psi\rangle = \sum_{x=0}^{2^n-1} a_x |U_x\rangle$$

$$\text{Hadamard Basis} \implies |v_x\rangle \in \psi = \sum_{x=0}^{2^n-1} a_x |U_x\rangle$$

$$\langle\psi|X_i|\psi\rangle = \left(\sum_{y=0}^{2^n-1} a_y^* |v_y\rangle\right) X_i \left(\sum_{x=0}^{2^n-1} a_x |v_x\rangle\right)$$

$$= \sum |a_x|^2 \langle v_x | X_i | v_x \rangle \leq 1$$

This is true since $\langle v_x | X_i | v_x \rangle$ is 1 or -1 based on the basis state of v_x . Thus the coefficients a_x which must add up to be less than or equal to one have the Hamiltonian to be minimized.

$$\langle\psi|X_i|\psi\rangle = 1 \implies \langle\psi|H_0|\psi\rangle = - \sum_{i=0}^{n-1} \langle\psi|X_i|\psi\rangle > -n$$

This implies that $\langle\psi|H_0|\psi\rangle > -n$ But we know that the value itself is $-n$ which implies that the state $|\psi_0\rangle$ is the minimum state. Q.E.D

6.3 Application

For quantum annealing, the Hamiltonian we wish to solve is

$$H_0 = \sum_{i=0}^{n-1} -X_i$$

$$|\psi_0\rangle = |+\rangle^{\otimes n}$$

$$H(t) = (1 - t/T)H_0 + t/TH_1, t \in [0, T]$$

$|\psi_T\rangle$ is the solution to the optimization problem, where T is the annealing time

D-Wave One exists such that 5000 Qubit Quantum Annealing Machine

7 Lecture 9: 3/22/24 Quantum Approximate Optimization Algorithm (QAOA)

For a discrete adiabatic QC:

$$H(t)|\psi\rangle = i\hbar\partial/\partial t|\psi\rangle$$

$$H(t) = A(t)H_0 + B(t)H_1$$

Applying discretization s.t.

$$t_k = k\Delta t, \Delta t = T/P$$

$$\implies H(t_k) = A(t_k)H_0 + B(t_k)H_1$$

$$|\psi_k\rangle \approx e^{j\Delta t H(t_k)} |\psi_{k-1}\rangle$$

$$\implies |\psi_P\rangle \approx \prod_{k=1}^P e^{j\Delta t (A(t_k)H_0 + B(t_k)H_1)} |\psi_0\rangle$$

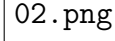
This is the solution to the Schrodinger equation (approximation). When we take Δt to be small, $e^{j\Delta t (A(t_k)H_0 + B(t_k)H_1)} \approx e^{j\Delta t A(t_k)H_0} \cdot e^{j\Delta t B(t_k)H_1}$

$$\therefore \psi_p \approx (\prod_{k=1}^P e^{j\beta_k H_0} e^{j\gamma_k H_1}) |\psi_0\rangle \equiv |\beta, \gamma\rangle$$

Where β and γ represent $e^{j\Delta t A(t_k)H_0} \cdot e^{j\Delta t B(t_k)H_1}$ respectively. The QAOA Circuit will look like the following

Thus we wish to minimize the following:

$$\min \langle \psi | H_1 | \psi_p \rangle = \langle \gamma, \beta \rangle$$



7.1 QAOA Hybrid Algorithm

- Classical: Initialize $\gamma\beta$
- Quantum: Prepare state $|\beta\gamma\rangle$
- measure $|\psi_p\rangle = |\beta\gamma\rangle$
- Estimate $E(\beta, \gamma)$
- If Solution found, then stop
- if solution not found, update $\beta\gamma$

7.2 QAOA Circuit

$$H_0 = -\sum_{i=0}^{n-1} X_i$$

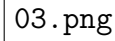
$$H_1 = -\sum_{i \neq j}^{n-1} J_{ij} Z_{ij} - \sum_i h_i Z_i$$

$$|\psi_0\rangle = |+\rangle^{\otimes n} = H^{\otimes n}|0\rangle$$

$$e^{j\beta_k H_0} = e^{-j\beta_k \sum_{i=0}^{n-1} X_i}$$

$$= \prod_{i=0}^{n-1} e^{-j\beta_k X_i}$$

From here, we apply the $R_x(2\beta_k)$ to the i-th qubit



$$\implies e^{j\gamma_k H_1} = \Pi_{i \neq j} e^{-j\gamma_k J_{ij} Z_{ij}} \Pi_i e^{-j\gamma_k h_i Z_i}$$

where $\Pi_i e^{-j\gamma_k h_i Z_i}$ represents the $R_z(2\gamma_k h)k$ gate.

Recall that

$$Z \otimes Z = |00\rangle\langle 00| + |11\rangle\langle 11| - |01\rangle\langle 01| - |10\rangle\langle 10|$$

Furthermore we have that

$$f(\mathbf{A}) = \sum \lambda_k \langle \lambda_k | | \lambda_k \rangle$$

which implies that

$$\exp(-j\theta/2 \cdot Z \otimes Z) = e^{-j\theta/2}(|00\rangle\langle 00| + |11\rangle\langle 11|) + e^{j\theta/2}(|01\rangle\langle 01| + |10\rangle\langle 10|)$$

This is the parity of the qubits and we can add the parities s.t. $|x_0 x_1\rangle = e^{(2y-1)\theta/2} |x_1 x_0\rangle$ Let $y = x_1 \oplus x_2$

Recall: $R_z(\theta)|y\rangle$ and the definition of it.

The circuit we will achieve is the following; $x_1 \otimes x_2 \implies |y\rangle \otimes |x_o\rangle \implies R_z(\theta)|y\rangle \otimes |x_o\rangle \implies e^{(2y-1)\theta/2} |x_1\rangle \otimes |x_0\rangle$

The circuit is then repeated p times, thus we have prepared the state properly.

7.3 Estimation Energy

We will represent the estimation energy as $E(\gamma, \beta) = \langle \beta_1 \gamma | H_1 | \beta_1 \gamma \rangle$

$$H_0 = - \sum_{i=0}^{n=1} J_{ij} Z_{ij} - \sum_i h_i Z_i$$

Recall that any computational state x is an eigenvector of Z_i, Z_{ij} w/ eigenvalue of ± 1 .

7.3.1 Example

$$H_1 = 3Z_{02} - Z_{01} + 2Z_0$$

Let $|x\rangle = |101\rangle$

$$\implies \langle x|H_1|x\rangle = 3(1) - 1(-1) + 2(-1) = 2$$

Thus we have that $\langle x|Z_i|x\rangle = 1$ if $x_i = 0$ and 0 if $x_i = 1$. We also have that $\langle x|Z_{ij}|x\rangle = 1$ if $x_i = x_j$ and -1 if $x_i \neq x_j$

Thus we can write down a minimization problem and a qubit string as a combination of $\langle x|Z_{ij}|x\rangle$ where for the computational vectors we can calculate the energy.

From here we can prepare M copies of the state ψ_p and we can measure the states. The energy corresponding to $E(\beta, \gamma) = \langle x|H_i|x\rangle$ which we can approximate to

$$\sum_x \frac{mx}{M} \langle x|H_1|x\rangle$$

7.4 Parameter training based on gradient descent

We wish to minimize $V(\theta) = \langle \psi_o | H | \psi_o \rangle$ where we wish to go down the gradient via the following:

7.4.1 Vanilla GD

where $O \implies O - \gamma \nabla_\theta V(\theta)$ where γ is the step size parameter.

7.4.2 Stochastic Gradient Descent

replace $\nabla_\theta V(\theta)$ by its simple estimate using N shots represent $\theta = (\theta_0, \dots, \theta_{n-1})^T \implies \nabla_\theta V(\theta) = (\partial V(\theta)/\partial \theta_0, \dots, \partial V(\theta)/\partial \theta_{n-1})^T$

Note: Back propagation is impossible here due to the fact that we cannot see the previous states of the qubits since they are in superposition.

We can approximate the partial derivative s.t. $\partial V(\theta) \approx$ to the limit definition of the partial derivative

$$[\nabla V(\theta)] = \frac{V(+\epsilon e_i) - V(\theta - \epsilon e_i)}{2\epsilon}$$

for $\epsilon > 0$ where ϵ is small. The realization is that this method takes $2Ns \cdot m$ measurements to estimate the gradient, thus implying expensive measurement. A better method is the SPSA method.

7.4.3 Simultaneous Perturbation Stochastic Approximation

let Δ be a random m-dimensional perturbation vector s.t. $\Delta \in [+1, -1]^n$ where we have that $[\nabla V(\theta)] = \frac{V(\theta+\epsilon\Delta)-V(\theta-\epsilon\Delta)}{2\epsilon\Delta}$. Thus there only needs to be $2Ns$ measurements to calculate the gradient.

7.4.4 Quantum Natural Gradient

We wish to create a quantum Fischer information matrix where

$$\theta \implies \theta - \gamma g^{-1}(\theta) \nabla_{\theta} V(\theta)$$

which is a much more costly method in terms of measurement

In general for the QAOA hybrid algorithm, we wish to know whether or not we have performed gradient descent or not.

8 Lecture 3/29/24

Ising Model:

$$f(z) = - \sum_{i \neq j} J_{ij} z_i z_j - \sum_i h_i z_i, z_i \in \{+1, -1\}$$

Hamiltonian:

$$H = - \sum_{i \neq j} J_{ij} Z_{ij} - \sum_i h_i Z_i$$

8.1 Warm Start QAOA

The goal of War Start QAOA is to relax the original QUBO s.t. $x_i \in [0, 1]$. We wish to solve the relaxed problem using a classical continuous optimizer - semidefinite program (SDP) denoted as solutions $\hat{x} \in [0, 1]$. We wish to set H_0 which is the initial Hamiltonian to $\hat{x}_i$ and let $\theta_i = 2\sin^{-1}(\hat{x}^i)$.

Then we let $H_0 = - \sum_{i=0}^{n-1} U_i$ where $H_i = I \otimes \dots \otimes I \otimes M - i \otimes \dots \otimes I$ where M_i is x at the i th position. Additionally, $M_i \equiv R_y(\theta_i)|0\rangle\langle 0|R_y(-\theta_i) - R_y(\theta_i)|1\rangle\langle 1|R_y(-\theta_i)$

Warm Start QAOA Circuit

8.2 High-orderbinary optimization (HOB0)

Ising model has terms: $z_{ij}, j \geq 2$. The Hamiltonian will be denoted be Z_{ij} . The variational principle still holds

9 Lecture 4/5/2024

VQE: Find the ground state of a given Hamiltonian:

$$H = \sum_l a_l G_l$$

which is denoted as the observer. Where a_l is an real number and G_l is a Pauli string tensor product of X, I, Y, Z

The ground state is denoted as

$$|\psi^*\rangle = \arg \min \langle H \rangle = \arg \min \langle \psi | H | \psi \rangle$$

.

The solution of this is going to be $|\psi^*\rangle = |\lambda_k\rangle$

Proof We have that

$$\begin{aligned} H &= \sum_{k=0}^{2^n-1} \lambda_k |\lambda_k\rangle \langle \lambda_k| \implies \langle \psi | H | \psi \rangle = \sum_k \lambda_k |\langle \psi | \lambda_k \rangle|^2 \\ &\geq \lambda_{min} \sum_k |\langle \psi | \lambda_k \rangle|^2 = \lambda_{min} \end{aligned}$$

where $\psi^2 = 1$. But $\langle \lambda_{min} | H | \lambda_{min} \rangle = \lambda_{min}$

Note: eigenvalues of $G_l = +1, -1$ and eigenvectors of G_l is the tensor product of $|0\rangle, |1\rangle, |+\rangle, |-\rangle, |1+j\rangle, |1-j\rangle$

Ex: $G_l = I \otimes X \otimes Y \otimes Z, n = 4$ One eigenvector is going to be $1 \otimes - \otimes 1 + j \otimes 0 \implies$ the eigenvalue is going to be 1.

9.1 For optimization: QUBO or HOBQ

G_l has only I or Z for QAoA. This is a lucky case because we will only have the tensor products of $|0\rangle$ and $|1\rangle$ since these are the eigenvectors of I and Z . This implies that the eigenvectors of G_l are going to be the computational basis.

For the ground state denoted as $|\psi^*\rangle$, this is going to be the computational basis due to the variational principle.

We usually use a classical circuit to solve for $\langle\psi_\theta|H|\psi_\theta\rangle$ and then we want to measure the value of ψ_θ and further do gradient descent. The question we thus are asking is how to estimate $\langle H \rangle_\psi$ based on measuring ψ in the computational basis

Let

$$|\psi\rangle = \sum_x b_x |x\rangle \implies \langle H \rangle_{|\psi\rangle} = \sum_l a_l \langle\psi|G_l|\psi\rangle$$

This term is going to become $\sum_x |b_x|^2 \langle x|G_l|x\rangle$ which is going to be either 1 or -1.

Example: $G_l = I \otimes Z \otimes I \otimes Z \otimes Z$ and $|x\rangle = |01011\rangle$ thus we are going to get that $\lambda_x = 1(-1)1(-1)(-1) = -1$

If we measure ψ in the computation basis with M shots, we get that $\sum_x m_x = M$. Therefore we can write the expectation value as

$$\langle H \rangle_\psi \approx \sum_l a_l \sum_x \frac{m_x}{M} \langle x|G_l|x\rangle$$

Thus the key is to calculate the objective value. The different G_l share the same eigenvectors and we can then calculate the expected value of every term and thus we can our solution. As we get closer to our optimal solution, we will see a larger spike in one of the expectation values.

9.2 Optimization for General VQE

Now, G_l has four terms: I, X, Y, Z We can write the eigen decomposition as

$$G = \sum_{\lambda=0}^{2^n-1} \lambda |\lambda\rangle \langle \lambda|$$

$$G = \sum_{x=0}^{2^n-1} \lambda |\langle x | G_l | x \rangle|$$

where λ_x is going to be 1 or -1 and the other term represents the tensor products of the different basis states $(0, 1), (+, -), \dots, (j, -j)$

We have $\lambda_x = U_g |x\rangle$ where $|x\rangle$ is the computational basis

Example;

$$\begin{aligned} G &= I \otimes X \otimes Y \otimes Z \\ |\lambda_x\rangle &= 0 \otimes + \otimes -k \otimes 1 \\ &= (I \otimes H \otimes SH \otimes I) |0011\rangle \\ \lambda_x &= 1(1)(-1)(-1) = 1 \end{aligned}$$

9.2.1 Exam Problem

Now we have that

$$\langle \psi | G | \psi \rangle = \sum_x \lambda_x |\langle \lambda_x | \psi \rangle|^2$$

Now let $|\psi\rangle = \sum_x c_x |\lambda_x\rangle \implies U_G^* |\psi\rangle = \sum_x c_x |x\rangle$. From here we get that

$$\begin{aligned} \langle H \rangle_{|\psi\rangle} &= \sum_l a_l \langle \psi | G_l | \psi \rangle \\ \langle H \rangle_{|\psi\rangle} &= \sum_l a_l \sum_x \lambda_{l,x} |\langle x | U_{G,l}^* | \psi \rangle|^2 \end{aligned}$$

For $l = 1, \dots, L$.

insert photo

We will first apply $V(\theta)$ on $|0\rangle$ and then we will get the state $|\psi\rangle$, we then apply $U_{g,l}^*$ and we then measure. For each outcome x , we get that

$$\lambda_{x,l} \in (1, -1) \implies \langle G_l \rangle_{|\psi\rangle} \approx \sum_x \lambda_{l,x} \frac{m}{M}$$

Differences between QAOA and VQE VQE is going to be for a general circuit while QAOA is going to be for a specific circuit with β and γ terms.

These are going to be the same algorithms but the circuits are going to be different.

In general VQE can deal with pauli matrices while QAOA only deals with X and Z .

9.3 Finding Excited States

In physics, we wish to find different excited states other than the minimum energy state. Let us find the following states $\lambda_0 < \lambda_1 < \lambda_2$. We will run VQE to find $\lambda_0 = \min \langle \psi | H | \psi \rangle$.

From here, we wish to find λ_1 . We will get the following:

$$H' = H + c \cdot |\lambda_0\rangle\langle\lambda_0|$$

and the eigenvectors are still going to be λ_j .

From here we will get that

$$H'|\lambda_0\rangle = (\lambda_0 + c)|\lambda_0\rangle$$

If $\lambda_1 < \lambda_0 + c \implies c > \lambda_1 - \lambda_0$. Then the ground state of $H' = |\lambda_1\rangle$ by running VQE.

We know that $|\lambda_0\rangle\langle\lambda_0|$ is not a pauli string, thus we must use a different way to calculate the Hamiltonian. Recall the following:

$$\begin{aligned}\lambda_0 &= V(\theta_0)|0\rangle \text{ ground state} \\ \psi &= V(\theta)|0\rangle \text{ excited state} \\ &\implies |\langle\lambda_0|\psi\rangle|^2\end{aligned}$$

is going to be simply equal to the probability of obtaining $|0\rangle$ by measuring $V(\theta_0)^*V(\theta)|0\rangle$

[image here]

Note: We do not know the value of c and thus must take it to be larger value than λ_1 simply by guessing.

9.4 Chapter 4: Quantum Machine Learning

We can consider the most simple machine learnign tasks: binary function to be learned

$$f : R^n \rightarrow [0, 1] \text{ or } [-1, 1]$$

We will have a training data set $S = \{(x^i, y^i), i = 1, \dots, k\}$ for x^i is the feature vector and y_i is the label.

[insert diagram]

9.5 Quantum Neural Networks (QNN)

The general structure of a classical neural network will ahve layers in which each layer has nodes. Each node is goign to be connected every other node in the next layer, creating a nural network between each subsequent layer. We will denote each layer as x^l, x^{l+1} and so on. Eventually, we will get a final node that we call z and perform a final calculation σ to get our final value $\hat{y} = p(y = 1|x)$

Fast Forward Neural Networks We denote the input as $x \in R^n$ with hidden layers from $l = 1, \dots, L - 1$. Additionally, we have a linear sublayer which we denote as $z^l = W^l x^{l-1} + b^l$ where $W^l \in R^{n \times n}$ and $b^l \in R^{n \times 1}$. We additionally have a nonlinear sublayer denoted as $x^l = g(z^l) \in R^n$. We then have out output layer which is going to be denoted as $z = \omega^L x^{L-1} + b^L, \omega^L \in R^{1 \times n}, b^L \in R$. Our final nonlinear sublayer we will get that $\hat{y} = \sigma(z) \in [0, 1]$ where σ is the output function for a binary classification.

The output function is going to be denoted as

$$\sigma(z) = \frac{1}{1 + \exp(-z)} \in (0, 1)$$

which is going to be our output function where $-\sigma(z)\sigma(-z)$ which is an odd function.

[insert graph]

Activation function: rectified in linear units (ReLU)

$$g(z) = z^+ = z, z \geq 0; 0, z < 0$$

[insert graph]

The problem with this is if we get to the value of zero in which we can no longer do gradient decent, i.e. $z < 0$. To generalize we are going to have that $g(z) = z^+ + \alpha z^-$ where $z^- = z, z < 0; 0, z \geq 0$. For $\alpha = -1$, we get that $g(z) = z$ and for α is fixed as a small number we get that $\alpha = 0.01 \rightarrow$ leaky ReLU. For a parametric ReLU, α is a learnable parameter

9.5.1 Loss Function and Regularization

Model parameters to be determined where θ is going to be a set $(W^1, b^1), \dots, (W^{L-1}, b^{L-1})$ and we are going to have the cost function to be $\theta^* = \arg \min J(\theta)$ where J is going to be the loss and regularization together which should represent the training data and structural constraints.

Loss Through Maximum Likelihood:

$$L(\theta) = -E_{(x,y)P_{data}} \log P_{model}(y|x) \approx - \sum_{i=1}^k \log P_{model}(y_i|x_i)$$

which the last term represents $\sigma(z)$.

This implies that the $-\log P_{model}(y|x)$ is going to be equal to $-\log \sigma(z), y = 1; -\log(1 - \sigma(z)), y = 0$ which is the same as representing this as

$$-\log[\sigma((2y - 1)z)]$$

which represents the cost function. By substituting the training data into the $(y_i|x_i)$ we can get the total cost function

Regularization This intends to reduce generalization error but NOT training error, using parameter norm penalties.

$$J(\theta) = L(\theta) + \alpha \Omega(\theta)$$

From this, we can have L^2 regularization

$$\Omega(\theta) = \frac{1}{2} \sum_i \omega_i^2 \rightarrow$$

make ω_i smaller. We can also have L^1 regularization such that

$$\Omega(\theta) = \sum_i |\omega_i| \rightarrow$$

sparsify the network. From here, we wish to get the optimal weights and thus must use gradient descent to do so.

9.5.2 Gradient Based Training

$\theta \rightarrow \theta - \epsilon \nabla_{\theta} \hat{J}(\theta)$ which the last term is the sample estimate of the loss.

From here, instead of using all training sample at each iteration, we create a sample minibatch of m samples from the training set. We are going to have multiple methods to do so:

Moment Method Gradient Using an exponentially decaying moving average of past where $v \leftarrow \alpha v - \epsilon \hat{g}$ and $\theta \leftarrow \theta + v$

1st moment:

$$s \leftarrow \rho_1 s + (1 - \rho_1) \hat{g}, \hat{S} = \frac{s}{1 - \rho_1^t}$$

for t iterations . And then for the 2nd moment we will simply have that :

$$r \leftarrow \rho_2 r + (1 - \rho_2) \hat{g} \cdot \hat{g}$$

$$\hat{r} \leftarrow \frac{r}{1 - \rho_2^t}$$

RMS prop Alg Scale the learning rates of all parameters based on the history of gradients. We have a vector

$$r \leftarrow \rho r + (1 - \rho) \hat{g} \cdot \hat{g}$$

$$\Delta \theta = -\frac{\epsilon}{\sqrt{\delta + r}} \cdot \hat{g}, \theta \leftarrow \theta + \Delta \theta$$

Adam - RMS prop + Moment In order for efficient calculation of the gradient, we will be using back propagation.

10 Lecture 4/12/2024

10.1 Variational Quantum Classifier

The Variational Quantum Classifier is going to be made of 3 parts: feature map, Ansatz, and projective measurement.

Recall that the cross-entropy loss for a classical neural network is going to be

$$L(\theta) = \sum_i l_i = \sum_i l_0(x, y), l_0 = -\log P_\theta(y|x)$$

QNN Can also use linear loss where $l_0 = 1 - P_\theta(y|x)$. Now given θ how do we estimate $L(\theta)$ for $i = 1, \dots, k$ (training sample), we wish to prepare M copies of $|\psi\rangle$ and measure each to get $z_i \in \{0, 1\}$. We get that

$$P_\theta \approx \frac{1}{M} \sum_{l=1}^M 1\{z_l = y_i\}$$

Cross Entropy Loss:

$$P_\theta \approx - \sum_{i=1}^k P_\theta(y_i|x_i)$$

Linear Loss:

$$P_\theta \approx - \sum_{i=1}^k \log P_\theta(y_i|x_i)$$

where z_i is the measurement.

The total number of shots for one evaluation of $L(\theta)$ is going to be Mk , The number of shots for one evaluation of $\nabla L(\theta)$ is going to be $2Mk$ for SPSA while the parametric shift rule is going to be $2Mk * \dim(\theta)$

From here, we can write the loss function in terms of observables where we recall that

$$P(y = 0) = \langle \psi | \pi_0 | \psi \rangle$$

Thus we get that

$$L(\theta) = - \sum_{i=1}^k \langle \pi \rangle_{|\psi_0(x_i)\rangle}$$

10.1.1 Feature Map (a.k.a. data encoding/loading)

Classical data $x \in R$ is mapped to a quantum state $|\psi(x)\rangle = U(x)|0\rangle$

Amplitude encoding maps a 2^n dimensional classical unit vector to n qubits thus we have

$$x \in R^{2^n}, |x| = 1 \rightarrow |\phi\rangle = \sum x_i |i\rangle, U(x) = \text{diag}(x_1, \dots, x_{2^n-1})$$

The advantages of this are that it requires only n qubits for 2^n elements. The disadvantages are that subsequent algorithms must operate on amplitudes, which is not efficient.

Angle encoding maps each element x_i to its own qubit.

$$x \in R^n \rightarrow |\phi(x)\rangle = \otimes_{i=0}^{n-1} R_Q(x_i)|0\rangle, Q \in X, Y, Z$$

If we have that $R_Q, Q = Z$, we must first apply the hadamard transform since applying the Z gate will only change the diagonals.

$$R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$$

10.1.2 Pauli Feature Map (with entanglement)

The purpose of this is to have advantages over classical approaches which need to have a map based on circuits that are hard to simulate classically.

In general, this consists of layers of Hadamard gates interleaved with entangling blocks. Within each entangling blocks are exponential functions of Pauli strings

Example: ZZ Feature Map We have a block of H gates and then $U(x)$ and then H gates and $U(x)$ and so on and so forth. Now lets say we have a feature map ZZ where

$$U(x) = \exp\left\{-j\left(\sum_{i=0}^{n-1} x_i Z_i + \sum_{i<j} (\pi - x_i)(\pi - x_j) Z_{ij}\right)\right\}$$

Recall the definition of $R_z(\theta)$ which is going to represent the first term of $U(x)$. The second term is going to be represented as $\exp(-j\frac{\theta}{2} Z \otimes Z)$

10.1.3 Parameterized Quantum Circuit (PQC)/Variational form/ansatz

Mean field ansatz use only single qubit gates where the output are separable states. Now we have that

$$R(o_1, o_2, o_3) = R_Q(o_1)R_{Q'}(o_2)R_Q(o_3), Q, Q' \in X, Y, Z, Q \neq Q'$$

where o represents the angle θ .

Hardware-efficient ansatz Are going to be single qubit gates with fixed entangling unitary

$$V(\theta) = V_L(\theta)V_{L-1}(\theta)\dots V_1(\theta)$$

$$V_l(\theta) = U_{ent}(R(\theta_{n-1}) \otimes \dots \otimes R(\theta_0))$$

where U_{ent} is composed of CNOT or CZ gates.

Example : Two local, NLocal, RealAmplitudes, Efficient SU2

10.1.4 Training PQC

We have different ways of training the PQC:

1. Numerical Difference
2. SPSA
3. Natural Gradient

Analytical gradient - parametric shift rule Assume each parameter θ_i contributes to the PQC ansatz through the gate of the form:

$$U_i(\theta) = \exp(j\theta/2P_i)$$

where P_i is a Pauli string. Recall that θ is a parameter and we can tune each parameter. This is satisfied by all ansatz seen so far except for the QAOA Circuit. From here we get the formula for the gradient:

$$\partial V(\theta)/\partial \theta_i = (V(\theta + \pi\theta_i) - V(\theta - \pi\theta_i))/2$$

Proof For each $k = 1, \dots, M$

$$U(O) = V(O_{-k} U_k(O_k) W(O_{-k})$$

where U_k has the same definition the previous U_i where P is a Pauli string. We end up with the fact that

$$\partial U_k(\theta) / \partial \theta_k = -1/2 P_k U_k(\theta)_k \implies U_k + 2\partial U_k / \partial \theta = (I - jP_k) U_k(\theta) \equiv S_k^+$$

where the term $I - jP_k = \sqrt{2} U_k(\pi/2)$ and we see that for the conjugate it will be equal to $\sqrt{2} U_k(\theta_k + \pi/2)$. *Note* the proof of the fact that

$$P = \pi_0 + \pi_1 = I$$

since the eigenvectors are orthogonal

We wil thus get that

$$\frac{\partial \langle \psi | \psi \rangle}{\partial \theta_k} = 1/4 \langle \phi_k | S_k^+ R_k S_k^+ | \phi_k \rangle$$

which is proved trivially :)

From here we get that

$$U(\theta) = \exp(-j\theta/2P) = \cos\theta/2I - j\sin(\theta/2P)$$

and thus we get that

$$U(\pi/2) = 1/\sqrt{2}(I - jP)$$

11 Lecture 4/19/24 Quantum Support Vector Machine (QSVM)

11.1 Classical Support Vector Machine (SVM)

There are two machine learning techniques: NN, DNN, SVM. SVM is based on rigorous theory of it.

11.1.1 Linear Binary Classification

We have a binary target function to be learned. $f : \mathbb{R}^2 \rightarrow [+1, -1]$. We will have a training data set $S = [(x, y) \dots (x_k, y_k)]$.

B: set of candidate functions. These class of functions are going to a set of linear classifiers where

$$B = x \rightarrow \arg \text{sign}(wx + b), w \in \mathbb{R}^d, b \in \mathbb{R}$$

This $wx + b$ is a hyper plane in a n dimensional space. Thus, we assign different values of +1 and -1 based on whether or not $\mathbb{R}^2$ maps to the correct values. This is for a simple case where there exists a simple graph that divides the two.

Hyperplane : hyperplane is a generalization of a two-dimensional plane in three-dimensional space to mathematical spaces of arbitrary dimension. Like a plane in space, a hyperplane is a flat hypersurface, a subspace whose dimension is one less than that of the ambient space

11.1.2 Separable Case

Assume that the training set S is linearly separable. That is, there exists $w, b \implies y_i(wx_i + b) \geq 0, i = 1 \dots k$ There is an infinite number of hyperplanes. Thus, we wish to define the hyperplane as the minimum distance between two different values. We will have that $w \cdot x + b = 0$ to be our hyperplane. Let us define $\rho(z)$ be defined as the distance away from the hyperplane where $\rho(z) = \frac{|wz + b|}{\|w\|}$.

We can now define the geometric margin of a linear classifier h for a training set S .

$$\rho_h = \min[\rho(x_1) \dots \rho(x_k)]$$

which is just the distance of the hyperplane to the two closest points of the training data in $\mathbb{R}^2$.

We can now form the SVM formulation. Our hyperplane will be the one that maximizes ρ_h . Thus we will denote

$$\rho^* = \max_{w, b} \rho_h = \max_{w, b} y_i \rho_h$$

From here we get that we simply want to maximize $\|w\|$ because the objective is invariant to scaling on (w, b) which implies that

$$\rho^* = \max \frac{1}{\|w\|}$$

Our primal fomrulation: $\min \frac{1}{2} \|w\|^2$ s.t. $y_i(wx_i + b) \geq 1, \dots, k$

Convex Problem: $\min f(x), x \in \mathbb{R}^n$ s.t. $g_i(x) \leq 0$. We will need to define a Lagrangian s.t.

$$L(x, a) = f(x) + \sum_{i=1}^k a_i g_i(x), a \geq 0$$

.

We will now define the condition for optimality. x^* is optimal if and only if there exists a^* s.t., $\frac{\partial L}{\partial x} = 0$ and $a_i^* g_i(x^*) = 0$ This implies that because $a^* > 0$ then $g_i(x^*) = 0$ which is an active constraint. When $a_i^* = 0$ then $g_i(x^*)$ is a inactive constraint

Dual Problem We wish to maximize the function for a . where $\max \theta(a)$ where $\theta(a) = \min_x L(x, a)$

The conclusion for the convex problem and the dual problem is that the $\theta(a^*) = f(x^*)$

Using this, let us consider

$$L(w, b, a) = \frac{\|w\|^2}{2} + \sum_{i=1}^k a_i [1 - y_i(wx + b)]$$

The KKT condition becomes $\dot{L}_w = w - \sum_{i=1}^k a_i y_i x_i = 0$ Looking at $\dot{L}_b = - \sum_{i=1}^k a_i y_i = 0$ which implies that the sum is equal to 0.

Vector x_i appears in w iff $a_i > 0 \implies w \cdot x_i + b = \pm 1$ because of the condition that $\dot{L}_a = 0$ which means that x_i is a support vector. The support vectors exist on the margin lines where the distance is going to be $\frac{1}{\|w\|}$

Now let us consider the solution to find a_i which is done by solving the dual problem. Recall the definition of $L(w, b, a)$. We will expand the second term to be

$$\sum a_i - \sum a_i y_i w x_i - \sum a_i b y_i$$

Because $\dot{L}_b = 0$, the third term is zero. And recalling the definition of w , we get that

$$\theta(a) = \sum_i a_i - 1/2 \sum_{i,j} a_i a_j y_i y_j (x_i \cdot x_j) \quad (1)$$

The dual problem is recalled to be $\max \theta(a), a_i \geq 0, \sum_{i=1}^k a_i y_i = 0$ Thus for any support vector x_i ,

$$b = y_j - w \cdot x_i \quad (2)$$

The SVM classifier is given by

$$h(x) = \text{sign}(w \cdot x + b) = \text{sign}\left(\sum a_i y_i (x_i \cdot x) + b\right) \quad (3)$$

The data here is only appearing inside an inner product

11.2 Kernel Methods

If linear separation is NOT possible, then we must for a non-linear decision boundary. This happens in two steps:

1. map the input space x to a higher dimensional feature space H
2. linearly separate $\phi(x)$ in H using SVM $\rightarrow$ in feature space, replace (x_i, x) with $K(x_i, x) = (\phi(x_i), \phi(x))$

11.2.1 Kernel

$K : X \times X \rightarrow \mathbb{R}$ s.t. for $x_1, x_2 \in X, K(x_1, x_2) = (\phi(x_1), \phi(x_2))$

Note that in classical SVM, we do not have to go to higher dimensions for the kernel method because we can directly use K through optimal constraints. (Mercers condition)

Example Polynomial Kernel : $K(x, z) = (x \cdot z + c)^d$. Suppose that this is two dimensional. Then we have that

$$k(x, z) = (x_1 z_1 + x_2 z_2 + c)^2 \quad (4)$$

Solving for (4) will give us the new vectors in the higher dimension. Thus we have $\mathbb{R}^2 \rightarrow \mathbb{R}^6$

Thus for examples, we can take the parity measurement in this polynomial kernel which allows us to form a linearly separable function.

11.2.2 Gaussian Kernel

We can replace each inner product (x, x') in the original SVM with the kernel function $K(x, x')$

11.3 QSVM

11.3.1 Quantum Kernel: Feature Map

1. classical data $x \in \mathbb{R}^n \rightarrow |\phi(x)\rangle \in \mathbb{C}^{2^n}$
2. $k(x_i, x_j)$ is the probability of $\phi(x)$

The ket of $\phi(x)$ is defined to be $U(x)|x\rangle$ which is the feature map circuit. Thus we will get that

$$K(x_i, x_j) = |\langle 0|U^*(x_i)U(x_j)|0\rangle|^2 \quad (5)$$

Therefore we can run the quantum circuit $U^*(x)U(x)$ on input $|0\rangle$ and measure the probability of the 0 output.

11.3.2 Quantum Kernel Estimation

Training stage: given the training data $(x_1, y_1) \dots (x_k, y_k)$ for $i = 1, \dots, k$ do: $K(x_i, x_i) = 1$ and then for $j = i+1, \dots, j$ apply the quantum circuit on $|0\rangle$ and then measure the probability of the output. And note $K(x_i, x_j) = k(x_j, x_i)$

Test Stage We can do the same thing except we don't need to normalize the value of $K(x_i, x_I) = 1$

Difference between Classical SVM Quantum feature maps are more expressive than the classical ones with the hopes of solving the kernel explicitly. Additionally, the quantum kernel estimation is noisy because of the shot noise.

11.4 Dimension Reduction: Principal Component Analysis

Because qubits are expensive, we wish to reduce the feature map to a low dimensional space. Data matrix $X[x_1, \dots, x_k] \in \mathbb{R}^{n \times k}$. We define an orthogonal projector $P = UU^T \in \mathbb{R}^{n \times n}$ with $U \in \mathbb{R}^{n \times d}$ containing d orthonormal vectors, $d < n$

For any $x \in \mathbb{R}^n$. $UU^T x$ projects x onto a d dimensional space. Therefore we can approximate $x \approx \sum_{l=1}^d a_l u_l$. We simply project x onto a lower dimensional space. Using the orthonormal basis u_l . By using this (principal component analysis), we can now use d qubits.

How do we find u_l ? These are going to be the eigenvalues of the eigenvectors of the correlation training data set.

11.4.1 PCA Solution

for ortho-correlation matrix $C = XX^T \in \mathbb{R}^{n \times n}$ and the eigen decomposition of $C = \sum_{i=1}^n \lambda_i u_i u_i^T$

Now we wish to take it so that we take the largest eigenvalues and $U = [u_1, \dots, u_d]$

04.png

05.png